

Caffeine Tracker

GitHub: <https://github.com/KL-Mithunvel/CAFFINE-TRACKER>

A local, single-user app for Windows 11 that logs every caffeinated drink you consume (in ml), computes the caffeine dose in mg, and shows a dashboard of graphs and statistics — including your max-intake day and how often you exceed the recommended 400 mg/day limit.

- **Quick entry** — one-tap presets (coffee, espresso, tea, Pepsi, Red Bull, Monster, ...) with typical serving sizes pre-filled.
- **Any other drink** — type its name and the app looks up its caffeine content online (Open Food Facts → bundled reference table → manual).
- **Offline-first** — no internet? The entry is saved immediately and the lookup is queued and retried automatically later.
- **Dashboard** — daily trend vs the limit, max intake date, days above the limit (count and %), averages, per-drink breakdown, time-of-day pattern, streaks.
- **PDF export** — one click in Settings produces a PDF report of your summary stats, per-drink breakdown and full entry log ([sample](#)).
- **Private** — everything lives in one SQLite file on your laptop.

Status: implemented. All 4 phases are built and covered by tests (pytest tests/ — 55 passing). See the documents below for design detail.

Mini-Project Submission

Student	Mithunvel K.L
Registration number	23BMH1029
Slot	G2+TG2
Project title	Caffeine Tracker
Report	project/PROJECT_REPORT.md · PDF: docs/Mithunvel_K_L_23BMH1029_G2-TG2.pdf

The report covers the **title, need, outcome and future perspectives**.

In short

- **Need:** caffeine content is rarely labelled and intake is spread over many drinks in different sizes, so people cannot tell how close they are to the recommended 400 mg/day. Existing apps need accounts or internet.
- **Outcome:** a private, offline-first desktop tracker that converts ml to mg automatically, looks up unknown drinks, and shows trends, streaks and limit breaches — 55 automated tests passing.
- **Future:** caffeine half-life / sleep warnings, limit reminders, barcode scanning, wearable data correlation, optional encrypted sync, installer.

Screenshots

Captured from the running app using synthetic data (scripts/seed_fake_data.py).

Entry

Caffeine Tracker

EntryDashboardManageSettings

Today's intake

268.4 mg / 400 mg limit

Quick add

Black Tea240 ml

Brewed Coffee240 ml

Coca-Cola350 ml

Espresso30 ml

Green Tea240 ml

Monster473 ml

Pepsi233 ml

Red Bull250 ml

Drink

Volume (ml)

When

e.g. Red Bull, or type any other :

dd-mm-yyyy --:--

Notes (optional)

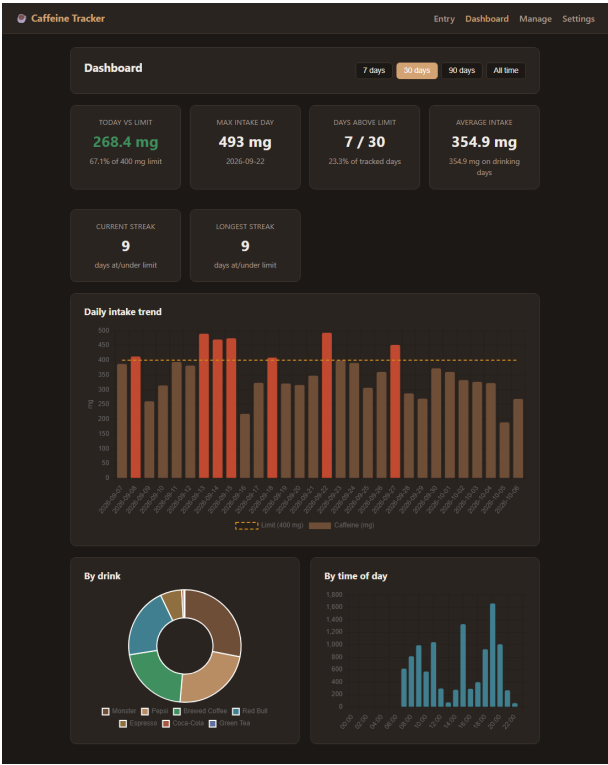
optional

Save entry

Logged today

Time	Drink	Volume	Caffeine	
18:00	Green Tea	240.0 ml	28.8 mg	✖
15:30	Red Bull	250.0 ml	80.0 mg	✖
11:00	Espresso	30.0 ml	63.6 mg	✖
08:15	Brewed Coffee	240.0 ml	96.0 mg	✖

Dashboard



Manage

Caffeine Tracker

EntryDashboardManageSettings

Manage entries

Date	Time	Drink	Volume (ml)	Caffeine (mg)	Notes	
2025-10-06	18:00	Green Tea	240.0	28.8		✖
2025-10-06	15:30	Red Bull	250.0	80.0	pre-lab	✖
2025-10-06	11:00	Espresso	30.0	63.6		✖
2025-10-06	08:15	Brewed Coffee	240.0	96.0	morning cup	✖
2025-10-05	20:00	Brewed Coffee	295.0	118.0		✖
2025-10-05	09:00	Coca-Cola	740.0	71.04		✖
2025-10-04	15:45	Pepsi	1075.0	115.03		✖
2025-10-04	11:00	Red Bull	410.0	131.2		✖
2025-10-04	09:30	Pepsi	710.0	75.97		✖
2025-10-03	19:45	Monster	530.0	179.14		✖
2025-10-03	18:30	Red Bull	460.0	147.2		✖
2025-10-02	15:45	Brewed Coffee	460.0	184.0		✖
2025-10-02	10:45	Monster	440.0	148.72		✖
2025-10-01	20:00	Brewed Coffee	580.0	232.0		✖
2025-10-01	10:30	Brewed Coffee	320.0	128.0		✖
2025-09-30	18:30	Monster	445.0	150.41		✖
2025-09-30	16:45	Red Bull	340.0	108.8		✖
2025-09-30	07:15	Monster	335.0	113.23		✖
2025-09-29	19:45	Pepsi	1130.0	120.91		✖
2025-09-29	16:30	Monster	180.0	60.84		✖
2025-09-29	08:45	Monster	260.0	87.88		✖
2025-09-28	22:00	Brewed Coffee	160.0	64.0		✖
2025-09-28	18:30	Monster	335.0	113.23		✖
2025-09-28	15:00	Monster	325.0	109.85		✖

Settings & PDF export

Caffeine Tracker

EntryDashboardManageSettings

Settings

Daily caffeine limit (mg)

400

Save

Default is 400 mg/day (FDA guidance for healthy adults). This value drives every "above limit" statistic on the dashboard.

Data Export

Export PDF Report

Generates a structured PDF with your summary statistics, per-drink breakdown, and full entry log.

Documents

Document	What it covers
project/PROJECT_REPORT.md	Mini-project report: title, need, outcome, future perspectives
project/REQUIREMENTS.md	What the app must do (FR/NFR), reference caffeine values
project/PROJECT_PLAN.md	Architecture decisions, repo layout, 4 build phases with success criteria
project/SCHEMA.md	Database design, data-type decisions, statistics queries
CLAUDE.md	Project brief for AI-assisted development sessions
TODO.md / Claude_log.md	Task tracker and session log

Tech stack

Python 3.12+ · Flask · SQLite · vanilla JS + Chart.js (bundled locally) · pywebview · YAML config · pytest.
Flask serves the app on 127.0.0.1 in a background thread; pywebview opens it in a native desktop window (no browser tab) — nothing leaves your machine except read-only caffeine lookups.

Running on Windows 11

Prerequisite: [Python 3.12+](#) with "Add python.exe to PATH" ticked during install.

```
git clone https://github.com/KL-Mithunvel/CAFFINE-TRACKER.git
cd CAFFINE-TRACKER
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python main.py
```

The app opens in its own window — not a browser tab. Double-clicking `start_tracker.bat` does all of the above for you (creates the venv, installs deps, runs `main.py`) with a visible console for logs/errors.

Desktop shortcut: run this once —

```
powershell -File scripts\create_desktop_shortcut.ps1
```

— to add a "Caffeine Tracker" shortcut to your Desktop that launches silently (no console window). If something goes wrong with a silent launch, check `logs\caffeine_tracker.log`.

Backup: copy `data\caffeine.db` anywhere — that one file is all your data.

Tests: `pytest tests/` (55 tests). **Demo data:** `python scripts\seed_fake_data.py` fills the dashboard with sample entries.

Project structure

<code>main.py</code>	entry point: Flask thread + pywebview window
<code>config.yaml</code>	port, DB path, lookup settings
<code>app/</code>	
<code>db.py</code>	SQLite connection, schema and seed data
<code>models.py</code>	CRUD for drinks / entries / settings
<code>stats.py</code>	all dashboard statistics
<code>lookup.py</code>	Open Food Facts -> reference JSON -> manual, retry worker
<code>pdf_export.py</code>	PDF report generator (ReportLab)
<code>routes.py</code>	thin HTTP layer
<code>templates/</code>	Jinja2 pages, vanilla JS, vendored Chart.js
<code>static/</code>	bundled offline caffeine table
<code>reference/</code>	
<code>scripts/</code>	desktop shortcut + demo-data seeder
<code>tests/</code>	pytest suite
<code>project/</code>	requirements, plan, schema, project report
<code>docs/</code>	screenshots, sample PDF, submission PDF

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